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Review

Carbon emission prediction models: A review

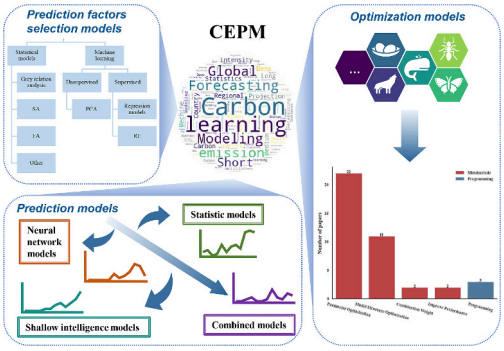
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HIGHLIGHTS

- We conduct a comprehensive review of 147 carbon emission prediction models.
- Examined models include prediction, optimization, and prediction factor selection.
- We analyze the advantages and disadvantages of each model.
- We compare the prediction performance of models in existing studies.
- We analyze the temporal and spatial distribution patterns of carbon emission prediction models.

GRAPHICAL ABSTRACT



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ABSTRACT

Amidst growing concerns over the greenhouse effect, especially its consequential impacts, establishing effective Carbon Emission Prediction Models (CEPMs) to comprehend and predict CO_2 emission trends is imperative for climate change mitigation. A review of 147 Carbon Emission Prediction Model (CEPM) studies revealed three predominant functions—prediction, optimization, and prediction factor selection. Statistical models, comprising 75 instances, were the most prevalent among prediction models, followed by neural network models at 21.8 %. The consistent rise in neural network model usage, particularly feedforward architectures, was observed from 2019 to 2022. A majority of CEPMs incorporated optimized approaches, with 94.4 % utilizing metaheuristic models. Parameter optimization was the primary focus, followed by structure optimization. Prediction factor selection models, employing Grey Relational Analysis (GRA) and Principal Component Analysis (PCA) for statistical and machine learning models, respectively, filtered factors effectively. Scrutinizing accuracy, pre-optimized CEPMs exhibited varied performance, Root Mean Square Error (RMSE) values spanned from 0.112 to 1635 Mt, while post-optimization led to a notable improvement, the minimum RMSE reached 0.0003 Mt, and

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the maximum was 95.14 Mt. Finally, we summarized the pros and cons of existing models, classified and counted the factors that influence carbon emissions, clarified the research objectives in CEPM and assessed the applied model evaluation methods and the spatial and temporal scales of existing research.

1. Introduction

Carbon dioxide (CO_2) has existed since the formation of the Earth. Plant photosynthesis and volcanic eruptions produce carbon dioxide in the environment (Rae et al., 2021). With the emergence of humans, carbon emissions are increasing (Gasser et al., 2015). This has increased the frequency and intensity of natural hazards. Some studies (Smith et al., 2016; van Hooidonk et al., 2016) have shown that by 2100, if emissions do not decrease, some tropical coastal areas will experience floods, fires, storms, and other disasters. To control carbon emissions and mitigate climate change, reducing carbon emissions has emerged as a worldwide objective for development (Murshed et al., 2023). In 1997, the 'Kyoto Protocol' stipulated carbon emission reduction as a legal obligation of developed countries (Duus-Otterström, 2023; Nations, U, 1998). More recently, the Paris Agreement, adopted in December 2015, led most countries to establish targets to mitigate global warming (Meinshausen et al., 2022; Nations, U, 2016). Setting goals and targets to reduce carbon emissions effectively is an essential part of climate action (Aboagye and Sharifi, 2023, 2024). An important initial step towards this is modeling carbon emissions (Han et al., 2023). Accordingly, in the realm of artificial intelligence, which is widely used today, carbon emission prediction has become a popular issue (Cabaneros et al., 2019). Precise forecasting of carbon emissions is essential to guide policy-makers effectively in mitigating the greenhouse effect (Hsu et al., 2022; Yin et al., 2022).

Several review studies on CEPM have been published in recent years. The following is a summary of the main findings from these reviews. Wang et al. (Wang et al., 2019a) discussed the historical factors influencing carbon emissions and projected future emissions within China's industrial sector. Grubb et al. (Grubb et al., 2015) reviewed 164 studies involving carbon emissions in China. They concluded that technological substitution is critical to reducing emissions to the lower end of the acceptable range. Dong et al. (Dong et al., 2018) elaborated on the current status of China's carbon emission peak, analyzing it from regional and sectoral perspectives and summarized various predictions. Hewage et al. (Abeydeera et al., 2019) reviewed carbon emissions literature from 1981 to 2019. They highlighted key research areas: carbon capture and storage, trend analysis for future carbon emission prediction models, evaluating carbon reduction performance, identifying emission reduction opportunities, and attaining zero carbon emission goals. In another study, Huang et al. (Huang et al., 2019a) examined 807 papers on China's sectoral carbon emissions. They found five common methods: extended environmental input-output analysis, index decomposition analysis, econometrics, evaluating carbon emission control efficiency, and simulation approaches.

Above studies provided a partial overview of methods applied in CEPM, yet they lacks comprehensiveness. Firstly, predicting carbon emissions often necessitates integrating various types of models. Simply classifying CEPM into statistical or artificial intelligence methods may lead to inaccurate descriptions. Secondly, the frequency and advantages and disadvantages of different CEPM remain unclear, impeding further innovation. Lastly, existing research has not thoroughly compared the predictive performance of CEPM. Therefore, a new systematic summary is required to provide detailed classification and in-depth analysis of CEPM.

To address these issues, this study conducted a comprehensive review of CEPM research from 2011 to August 2022, aiming to achieve the following objectives:

- Establish a classification framework for the three main models in CEPM: factor selection models, prediction models, and optimization models, elucidating the strengths and weaknesses of each.
- Classify and discuss the influencing factors, research objectives, and performance evaluation methods of CEPM.
- Conduct a comprehensive comparison of reviewed CEPM using the three main error methods: RMSE, MAE, and MAPE.
- Analyze the temporal and spatial distribution patterns of CEPM and discuss carbon emission trends in major countries.

This paper is organized as follows: Section 2 delineates the research methodology and elucidates the search strategy employed. In Section 3, we provide a comprehensive categorization and review of CEPM, highlighting its strengths and limitations. Within Section 3, we initially categorize predictive models in CEPM, followed by an examination of optimization models used for carbon emissions modeling, including programming and meta-heuristic analyses. Furthermore, the last part of Section 3 involves an analysis of reported prediction factor selection models. Section 4 engages in a discussion of CEPM, exploring the specifics of carbon emission models, including the classification of influencing factors, relevant statistical analyses, study targets, prediction model evaluation methods, considerations of time and spatial scales in various papers, future carbon emission trends in different countries, the limitations, and potential solutions for these approaches. Section 5 concludes the paper.

2. Materials and methods

2.1. Boundary definition of carbon emissions

Generally, carbon emissions refer to carbon dioxide emissions, which are also the primary contributor to global warming (Pachauri et al., 2014). As a result, this review does not include articles centered on other greenhouse gases. Furthermore, the concept of carbon intensity, which signifies the volume of carbon emissions produced per unit of GDP, has garnered increased scholarly attention in recent years and has thus been incorporated into this review (Acheampong and Boateng, 2019; Ye et al., 2018; Wang et al., 2018).

2.2. Search strategies

We followed a rigorous article screening method. First, we searched all published studies in the core database of the Web of Science to summarize the research trends of the most authoritative papers. The main search terms included different combinations of words such as 'carbon emission', 'prediction', 'forecasting', 'modeling', and 'machine learning'. Word list for literature search is presented in Table 1. The search process iterated until no further pertinent citations were acquired. Additionally, the reference lists of the chosen papers underwent

Table 1
Word list for literature search.

Method	Model	Objective	Geographical extent	Time horizon
Forecasting	Modeling	Carbon emission	Global	Short
Projection	Machine learning	Carbon intensity	Regional	Medium
	Deep learning		Country	Long
	Statistics			

analysis to identify crucial references. Ultimately, only literature published in English was included. The earliest CEPM research we indexed was from 2011, and the literature search continued until the day before the article was written. Therefore, the scope of the review papers was from 2011 to August 2022. Finally, 147 papers that met the established requirements were included in this study. The Online Supplementary Appendix I lists studies on CEPMs published between 2011 and 2022 that were used in this review. The literature search strategy and methodology for literature selection are illustrated in Fig. 1.

3. Results

3.1. Prediction models

After a detailed and comprehensive literature survey, as shown in Fig. 2, the prediction models used in CEPM were divided into four categories. We counted the main models used for analysis in each paper. The number of papers using statistical models was the largest (75 papers), with neural network models (31 papers), and shallow intelligent models in the field of machine learning (27 papers) being the second- and third-most common, respectively. A combined prediction model was defined as a combination of two or more prediction models, and these models were least common (20 papers) among the 147 papers reviewed.

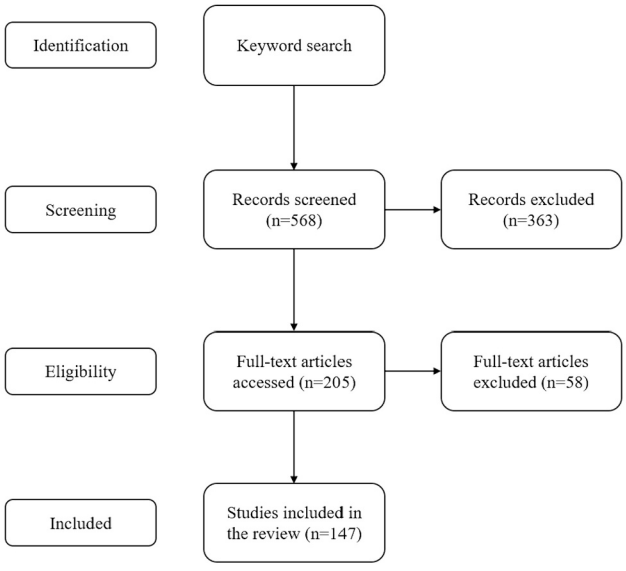
3.1.1. Statistical models

Fig. 2 shows the classification and quantitative statistics of statistical models for emission prediction. The grey model (GM) was implemented 43 times, and 7 papers applied the classic regression model. In addition, 19 papers used other statistical models. Among the GMs considered, GM (1, 1) was most used (26 papers), even more often than classic statistical methods. Classic regression methods include linear regression and panel regression, which were used in 5 and 2 papers, respectively.

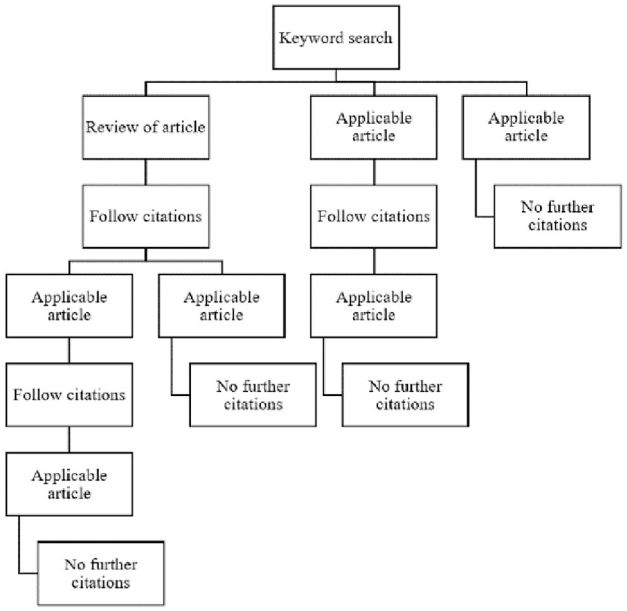
Regression analysis is an iterative process. In CEPM, two regression methods are used: panel regression (PR) and linear regression (LR). One reviewed paper used an LR approach. Shweta et al. (Singh and Kennedy, 2015) proposed a regression model to develop a tool for predicting carbon emissions in 3646 cities worldwide. In addition, four papers used PR. Song et al. (Wang et al., 2019b) used input–output analysis and PR to predict trade-related carbon emissions in China and Australia through 2025. The findings indicated a growth in net carbon export from China to Australia, ranging between 2.2 and 15.5 Mt of carbon. Yuan et al. (Yang et al., 2018) used dynamic panel data model to predict the carbon intensity in China. By 2020, the carbon intensity in China decreased by 33 % from the 2005 level. Zhao et al. (Zhao and Du, 2015) used PR to forecast the carbon emissions of 90 OECD countries and China. The results showed that CO_2 emissions are expected to be approximately 25 % lower than 2010 levels by 2050. Du et al. (Du et al., 2012) utilized PR to predict provincial carbon emissions in China spanning from 2010 to 2020.

In the papers studied, two univariate time series methods are used: Autoregressive integrated moving average (ARIMA) and Autoregressive distributed lag (ARDL) models. The general form of ARIMA, symbolized as ARIMA (p, d, q), involves 'p' indicating autoregressive order, 'd' representing difference order, and 'q' indicating moving average order (Kaur et al., 2023). The four reviewed papers use the ARIMA method. Li et al. (Li et al., 2020a) concluded, based on the ARIMA model's predictions, that China's anticipated peak emissions would reach 96.3 million tons by the year 2021. Malik et al. (Malik et al., 2020) employed the ARIMA model to forecast carbon dioxide emissions in Pakistan up to 2030. The model demonstrated good accuracy, with a generally low Mean Absolute Percentage Error (MAPE) value consistently below 10 %. Yang et al. (Yang and O'Connell, 2020) utilized the ARIMA model to forecast carbon emissions within the Shanghai aviation industry. Their findings revealed a projected continual increase in carbon emissions. Lin et al. (Lin and Agyeman, 2019) formulated an ARDL model to forecast carbon emissions in Ghana spanning from 2017 to 2030.

In 1982, Deng Julong (Deng, 1982) proposed the grey system theory, tailored for modeling systems with partial unknowns, small samples, and limited information. The grey prediction model, a pivotal component of this theory, finds applicability across diverse fields, including CEPM. With continuous applications and updates, many improved traditional grey models have been proposed and used for CEPM. A total of 43 of the studies reviewed applied GMs and their variants, accounting for approximately 29.1 % of all prediction models reviewed. Grey models can be divided into univariate and multivariate models, which



a.



b.

Fig. 1. (a) Approach for literature search and selection - adapted from Moher et al. (2009). (b) The methodology employed in selecting the reviewed papers.

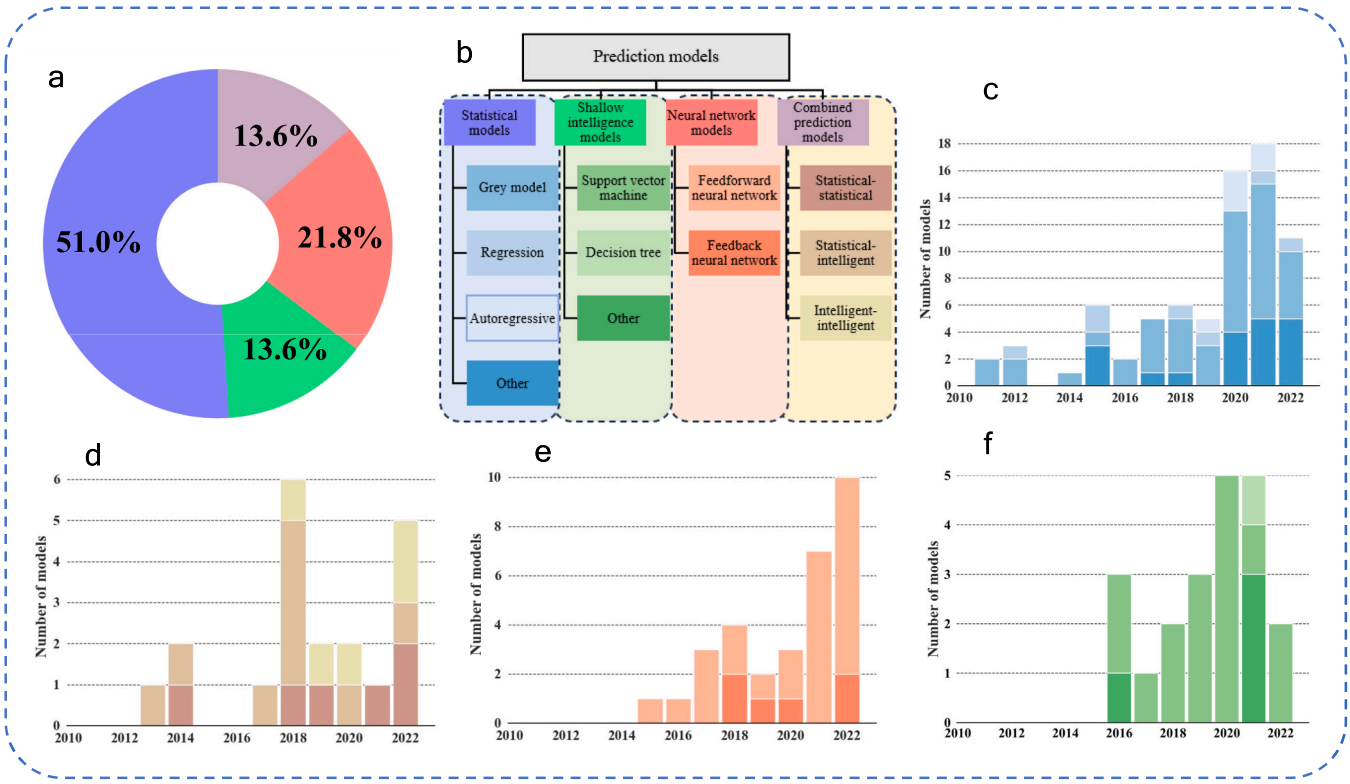


Fig. 2. (a) Proportion of the four prediction models among all models, with blue indicating statistical prediction models, green for shallow intelligent prediction models, red for neural network prediction models, and purple for combined prediction models. (b) Classification of prediction models. (c) Time series distribution of different statistical prediction models, color-coded as shown in (b). (d) Time series distribution of different combined prediction models. (e) Time series distribution of different neural network prediction models. (f) Time series distribution of different combined prediction models.

are denoted as GM (1, 1) and GM (1, N), respectively. Among them, a total of 26 papers used the univariate grey model, and 17 papers used the multivariate grey model. GM (1, 1) and GM (1, N) models are two basic models in grey theory (Xie and Wang, 2017). The number in the first bracket indicates the order of the grey differential equation, while the second number signifies the count of variables involved (Liu et al., 2016). This model aims to employ an Additive Generation Operator (AGO) on the initial data, constructing a new sequence based on exponential rules. Subsequently, by fitting this new data sequence with an exponential model, the Inverse Additive Generation Operator (IAGO) is applied to forecast a value for the original sequence (Liu and Yang, 2017). Moreover, the GM (1, 1) model has been used in predicting carbon emissions. (Yu and Xu, 2019; Yu et al., 2018; Li et al., 2018a). Various iterations of GMs have found application in CEPM. These models can be categorized into equigap grey models and non-equigap grey models based on their model spacing (Wu et al., 2020). These improved methods mainly include processes such as initial condition optimization, background value optimization, and the whitening of grey derivatives (Xie, 2022; Wang et al., 2011). In addition, GMs are characterized by strong generalization ability, so they can be easily combined with other models, such as the grey Verhulst model and nonlinear grey Bernoulli model (NGBM) (Duan and Luo, 2020; Pao et al., 2012).

The grey Verhulst model stands as one of the frequently employed models in predicting within grey systems. This method is particularly effective for data that exhibit a relatively unimodal trend (initially increasing or decreasing before reversing). In 2020 (Duan and Luo, 2020), the Verhulst (1, 1, β) model was introduced, leveraging the particle swarm optimization (PSO) algorithm to optimize the background variable β. The outcomes illustrated the superiority of this proposed model over traditional grey models in predicting carbon emissions from coal-related sources. Another advancement in 2018 (Wang and Li, 2019), introduced the non-equigap grey Verhulst model, utilizing the PSO algorithm to optimize its structural parameters. Jiang (Jiang et al., 2021a) proposed an enhanced grey multivariate Verhulst model (GMVM (1, N)). This model integrates the grey Verhulst (1, N) model with a residual correction model while preserving the original parameters. Additionally, the grey control parameters from GM (1, 1) were incorporated into the model to enhance prediction accuracy. Chen (Chen, 2008) coined the term "nonlinear grey Bernoulli model" (NGBM). The NGBM-OP model, introduced by Pao et al. (Pao et al., 2012), is distinguished by its robust predictive capacity, showcasing an MAPE (Mean Absolute Percentage Error) of under 6.3 %. Employing NGBM-OP, predictions were made for carbon emissions, energy trends, and outputs spanning 2009 to 2020.

3.1.2. *Shallow intelligent models*

We classify models with lower structural complexity and smaller parameter scales within intelligent models as shallow intelligent models. This classification aims to distinguish them from neural network models. In the 147 papers reviewed, there are two main shallow intelligent prediction models: support vector machine (SVM) and decision tree model. The SVM model can be further divided into LSSVM (least squares support vector machine) and SVM models, which are used in 9 and 7 papers, respectively. The decision tree model only appears once in our literature review. The classification and quantitative statistics of the said models are given in Fig. 2.

SVM stands as a robust classification and regression tool, rooted in the Vapnik-Chervonenkis (VC) dimension theory, and guided by the principle of structural risk minimization. These models are characterized by fast learning speed and good generalization ability (Ahmad et al., 2014; Chauhan et al., 2019). Wang et al. (Wang et al., 2020) compared backpropagation neural network (BPNN), Gaussian process regression (GPR), and SVM, among which the SVM yielded better accuracy than the other models. The hyperparameters of SVM highly affect

the accuracy of modeling. The most important is C, which is the penalty coefficient, or the tolerance for error. A higher 'C' signifies increased intolerance to errors, potentially leading to overfitting. Conversely, smaller 'C' values may lead to underfitting. Additionally, excessively large, or small 'C' values can compromise the model's generalization ability. Therefore, many studies involving CEPM are based on optimized SVM models (Wen and Cao, 2020a; Wang et al., 2019c; Sutthichaimethee and Kubaha, 2018). In addition to PSO, optimization models are based on other metaheuristic methods, as will be discussed in the next section. The LSSVM was first proposed by Suykens et al. and is applied for problem-solving in pattern classification and function estimation tasks (Suykens and Vandewalle, 1999). A least squares linear system replaces the quadratic programming approach typically used in SVMs. This approach simplifies the calculational complexity and improves the operating speed of the model. Like that of artificial neural networks (ANN) and other intelligent algorithms, the performance of LSSVM models depends on the selected inputs and parameters (Sun and Zhang, 2020; Wen and Cao, 2020b). Zhu et al. (Zhu et al., 2019) predicted energy consumption and carbon intensity by introducing a new LSSVM with a mixed kernel function. The results indicated that the proposed model outperformed the original one in terms of accuracy. Sun et al. (Sun and Liu, 2016) proposed an LSSVM to predict the CO2 emissions of different industries in China. Their findings revealed that the LSSVM exhibited superior accuracy compared to BPNN, GM (1, 1), and logistic models.

GBDT (gradient boosting decision tree) is an ensemble algorithm based on a decision tree. This ensemble boosting method uses a classification and regression tree (CART) as the basic learner (Cheng et al., 2019). GBDT can be used to solve most regression problems by generating multiple weak prediction models and combining them, finally generating an optimal performance model in an iterative manner (Li et al., 2020b). Cui et al. (Cui et al., 2021) developed a GBDT prediction model for forecasting carbon emissions in China. They optimized the four parameters of the GBDT using the MWOA.

3.1.3. Neural network models

In the models reviewed, 32 papers applied different forms of neural networks. In the 147 CEPM studies reviewed, two types of neural networks emerged: feedforward neural networks (FNN), and feedback neural networks. The feedback neural networks are mainly RNN models. The feedforward neural networks can be divided into single-hidden-layer neural networks and multiple-hidden-layer neural networks based on the number of hidden layers. The classification and quantitative statistics of different types of neural networks are shown in Fig. 2.

ANN is a mathematical model algorithm that mimics the distributed, parallel information processing seen in animal behavior (Hubara et al., 2018). This network accomplishes information processing by adjusting the connections between numerous internal nodes. In an ANN network, each neuron receives input signals, processes them using an activation function, and generates output signals. The connections between neurons have weights, which determine the strength of influence of one neuron on another. During training, these weights are adjusted based on specified learning algorithms, such as backpropagation, to minimize the difference between the actual output and the expected output (Mocanu et al., 2018; Samek et al., 2021). Notably, Ghalandari et al. (Ghalandari et al., 2021) employed a Multiple-Hidden-Layer Artificial Neural Network, known as an MLP model, for predicting carbon emissions in the UK, Germany, Italy, and France. Their findings indicated that the MLP model exhibited lower error rates compared to a GMDH model.

NNAR (p, k), short for Nonlinear AutoRegressive Neural Network, is a type of feedforward neural network designed specifically for time series analysis. In NNAR (p, k), the input to the neural network consists of lagged values of the time series data. It comprises a single hidden layer with p nodes, each representing a lagged input, and k nodes in the output layer. This architecture allows the model to capture nonlinear relationships within the time series data (Daniyal et al., 2022). The model can be applied to original nonlinear data without any parameter restrictions required to ensure stability. Cristiana (Tudor, 2016) predicted carbon emissions in Bahrain using an automated prediction method. Seven different prediction models were considered. According to the RMSE metric, NNAR yielded the highest accuracy.

From 1986 to 1988, psychologists L.L. McClelland and D.E. Rumelhart proposed the famous multilayer neural network with an error backpropagation algorithm, namely, the BPNN (Rumelhart et al., 1986). In BPNN, carbon emission data flow forward through the network from the input layer to the output layer, while error messages propagate backward from the output layer to the hidden layer. During training, the network computes the error between the actual output and the expected output, then adjusts the connection weights between neurons to minimize this error. The backpropagation algorithm involves both forward propagation and backward propagation. During forward propagation, input data is fed through the network and outputs are computed. During backward propagation, errors propagate backward through the network, and gradient descent is used to optimize the adjustment of the connection weights accordingly (Lillicrap et al., 2020; Wright et al., 2022). Wen et al. (Wen and Yuan, 2020) used BPNN optimized with RF and PSO to forecast the CO2 emissions of the commercial sector of China. Zhou et al. (Zhou et al., 2017a) utilized an enhanced BPNN optimized by PSO to forecast carbon emissions within the thermal power industry in the Beijing-Tianjin-Hebei region of China. Their findings indicated that BPNN achieved an error rate of <6 %.

GRNN represents Generalized Regression Neural Network. Unlike traditional feedforward neural networks with layered organization of neurons, GRNN has a radial basis function (RBF) architecture (Al-Mahasneh and Anavatti, 2023). In GRNN, each neuron in the hidden layer is associated with a radial basis function, which computes the similarity between the input data and prototype vectors. The output of each neuron is weighted by this similarity measure. Training of GRNN involves prototype selection and parameter estimation. Prototype selection entails determining prototype vectors from the training data, which represent the distribution of input data in the feature space. Parameter estimation involves estimating the width of the radial basis functions and the weights associated with each neuron. (Zhang et al., 2023). In 2020, Niu et al. proposed an improved GRNN-based CEPM for predicting TCE and CEI in China until 2040. (Niu et al., 2020).

The fundamental architecture of a CNN consists of distinct layers: an input layer, a convolutional layer, a pooling layer, a fully connected layer, and an output layer (Zhou, 2020; Gu et al., 2018). Typically, parameters for the convolutional and pooling layers are set through iterative experimentation. Neurons at the output feature surface of the convolutional layer are locally connected to the input of that layer. The corresponding connection weight and local input weight are calculated, added to the bias value, and used to determine the neuron's input value. This process allows CNNs to automatically learn hierarchies of features from input carbon emission data (Basha et al., 2020; Kiranyaz et al., 2021). In CEPM, Hien et al. (Hien and Kor, 2022) utilized a CNN model to forecast Canadian fuel-related carbon emissions. Their findings highlighted the CNN model's superior stability compared to a regression model in this domain.

All parameters of the traditional FNN need to be adjusted, so there are dependencies (weights and deviations) among different parameter layers. The gradient descent method is often applied to various FNNs (Chen et al., 2019). However, these methods based on gradient descent usually take a long time to run and easily fall to local optima. Unlike the traditional FNN, ELMs are new learning algorithms based on a single-hidden layer-FNN, which is used to randomly select hidden nodes and analyze the subsequent output weights (Ma and Dai, 2016). In the field of CEPM, Wei et al. (Sun et al., 2018) proposed an ELM model based on RF selection prediction factors and moth flame algorithm optimization. Their results demonstrated its superior prediction accuracy over BPNNs in similar conditions. Additionally, Sun et al. (Sun et al., 2017) utilized PSO to optimize input weights and deviation thresholds within an ELM.

Subsequently, this optimized ELM was employed to forecast carbon emissions in Hebei, China.

Unlike feedforward neural networks above, RNN is a typical feedback neural network. Their cyclic structure enables the transmission of information from preceding moments to subsequent ones, facilitating specific actions at each moment (Sherstinsky, 2020). The hidden layer of a RNN contains recurrent connections, enabling the network to capture temporal dependencies in carbon emission data. At each time step, the hidden layer receives input from both the current input data and the previous hidden state, allowing it to retain information over time. The output layer generates predictions or outputs based on the information learned by the hidden layer (Sherstinsky, 2020). Building upon this concept, Mason et al. (Mason et al., 2018) proposed an RNN model in which the parameters were optimized with the CMA-ES evolutionary algorithm. Finally, 2.5-hour-ahead carbon emissions and energy consumption in Ireland were successfully predicted.

An RNN's hidden state relies on the previous moment's hidden layer and the current input, lacking long-term memory advantage. Hence, Sepp Hochreiter et al. introduced LSTM as an enhanced RNN model (Hochreiter and Schmidhuber, 1997). LSTM is a type of RNN designed to handle sequential data and capture long-term dependencies. Unlike RNNs, LSTM addresses the vanishing gradient problem by introducing memory units and three gate mechanisms, input gate, forget gate, and output gate. The memory units of LSTM maintain a constant state over time, allowing them to retain information in long sequences. The input gate controls the flow of new information into the units, while the forget gate regulates the retention of previous state information. The output gate determines which parts of the unit's state are output to the next layer. These gate mechanisms controlled by activation functions enable LSTM to selectively update and forget information based on input and past states, effectively capturing complex temporal dependencies (Van Houdt et al., 2020). Bismark et al. (Ameyaw et al., 2020) predicted carbon emissions in West Africa from 2015 to 2030 by using a bidirectional long short-term memory (BiLSTM) model. Huang et al. (Huang et al., 2019b) applied LSTM for predictions in China, revealing an anticipated 30 % decrease in carbon emissions per unit of GDP between 2015 and 2020. Similarly, Bismark et al. (Ameyaw and Yao, 2018) utilized BiLSTM to forecast carbon emissions in Ghana, Nigeria, Burkina Faso, Senegal, and Benin spanning from 2015 to 2020.

3.1.4. Combined prediction models

Combined prediction models hold significance in CEPM. They involve crafting multiple individual prediction models and integrating them strategically to create a unified model (Tascikaraoglu and Uzunoglu, 2014; Wang and Srinivasan, 2017). The reviewed combined models can be divided into three groups: statistical-statistical models, statistical-intelligent models, and intelligent-intelligent models, as shown in Fig. 2. The intelligent models include shallow intelligent prediction models and neural network prediction models. Based on our review, combined statistical-intelligent prediction models are the most common among researchers (9 papers). The comparative analysis of combined models is shown in the Online Supplementary Appendices (II, and III).

Statistical-statistical models involve combining multiple classic statistical prediction models to enhance forecasting. Its objective is to amalgamate the strengths of various statistical models, thereby improving prediction accuracy and stability. These models, based on statistical theory, encompass linear regression, polynomial regression, exponential smoothing, among others. By integrating the predictions of multiple foundational models, Statistical-statistical models mitigates forecasting errors and heightens precision. Additionally, it counters limitations in individual models, thus bolstering overall forecast stability. Nonetheless, determining combination weights relies on experimentation and experience, and for intricate problems like nonlinearity or non-stationarity, these models may encounter limitations. Wang et al. (Wang et al., 2019d) established a combined prediction model based on the Kaya identity, the logarithmic mean Divisia index (LMDI) model, and a scenario analysis method. Their combined model predicted China's CO_2 emissions to reach 112.89 million tons by 2030. Cui et al. (Cui et al., 2018) predicted carbon dioxide emissions from China's power sector using hybrid PLS grey Markov model, foreseeing an increase to 51.029 million tons by 2025. Moreover, Meng et al. (Meng et al., 2014) combined the GM (1, 1) prediction equation with a linear model, creating a mixed equation. Their analysis, focusing on forecasting China's carbon emissions from 1992 to 2011, demonstrated that the hybrid model outperformed the traditional linear model and GM (1, 1) in accuracy.

Statistical-intelligent models merge classical statistical prediction models with intelligent models to enhance forecasting accuracy and stability. This approach aims to leverage the strengths of statistical and machine learning models. Within this framework, some foundational prediction models are grounded in statistical theory, while others stem from machine learning algorithms (e.g., support vector machines, neural networks). This approach combined the interpretability of statistical models and the ability of intelligent models to handle nonlinearity and high-dimensional data. In response to the random oscillation sequence (ROS) pattern observed in industrial carbon emission processes, Hu et al. (Hu and Lv, 2020) developed the ROGM-AFSA-GVM model. This model combined a ROS-based Grey Model (GM) with a General Vector Machine (GVM) optimized using the artificial fish swarm algorithm (AFSA). Li et al. (Li et al., 2018b) employed the GM (1, 1) predicted output as input for the SVM-ELM model to forecast carbon emissions in the Beijing-Tianjin-Hebei region, revealing potential emissions control below 96.9 million tons by 2030. Wang et al. (Wang et al., 2018) combined a GM and a GRNN model to predict China's primary energy consumption from 2017 to 2030, utilizing the GRA to allocate weights between the models. Zhao et al. (Zhao et al., 2018a) constructed a mixed frequency data sampling (MIDAS) regression model and a hybrid BPNN (MIDAS-BP) approach to predict CO2 emissions in the United States. The results showed that the prediction ability of MIDAS-BP is significantly better than MIDAS, and the other four individual prediction models. Zhou et al. (Zhou et al., 2017b) created a hybrid GNNM model, amalgamating GM and BPNN models for carbon emission prediction. Their findings demonstrated that the GNNM model offered improved predictions for carbon emissions, effectively capturing the nonlinear and nonstationary nature of carbon emissions.

Intelligent-intelligent models involve merging diverse shallow intelligent prediction models and neural network prediction models to capitalize on their respective strengths. By combining predictions from various intelligent models, this approach reduces prediction errors and improves precision. Leveraging the advantages of different machine learning models in handling nonlinearity, high-dimensional data, and capturing intricate relationships is the cornerstone of this approach. Yet, handling complex numerical features may lead to overfitting issues. Li (Li, 2020) constructed a KLS algorithm by combining Kalman filter (KF), LSTM and SVM to predict carbon emissions in China from 2015 to 2030. The results indicated that the carbon equivalent in China will continue to increase, peak in 2024, and then gradually decline. In addition, Acheampong et al. (Acheampong and Boateng, 2019) utilized an ANN-feedforward multilayer perceptron (FFMLP) model to project carbon emissions across Australia, Brazil, China, India, and United States. Their outcomes highlighted an exceedingly minimal prediction error between the forecasted and actual carbon emissions.

3.1.5. Summary of the prediction models

In statistical prediction models, regression models serve as concise and intuitive statistical tools with good interpretability, especially the linear regression. However, their adaptability to nonlinear relationships is limited, and when dealing with complex, high-dimensional carbon emission issues, regression models may struggle to comprehensively consider various influencing factors. Univariate time series methods, such as ARIMA, perform well in handling trend and seasonality in data

but may be influenced by fluctuations when applied to carbon emissions data affected by multiple factors. Grey models (GM), particularly GM (1, 1), are widely used in CEPM, constituting 29.1 % of all reviewed models. While GM excels in handling small sample sizes and incomplete information, it has notable drawbacks. GM often relies on assumptions about initial values during the modeling process, heavily dependent on the researcher's experience. Although enhanced grey models improve predictive accuracy, they introduce more complexity, requiring additional parameter tuning and optimization.

We conducted a survey on two shallow intelligent prediction models: Support Vector Machine and Decision Tree. SVM is characterized by its advantages, including a faster learning speed and robust generalization ability. However, its effectiveness is highly dependent on hyperparameters, particularly the penalty coefficient 'C.' LSSVM, an improved version of SVM, simplifies computations by replacing quadratic programming with least squares. While this simplification enhances computational efficiency, LSSVM carries the risk of overfitting, especially when the penalty coefficient is too high. On the other hand, Gradient Boosting Decision Tree (GBDT), as an ensemble model using decision trees as basic learners, demonstrates high accuracy. GBDT enhances precision by iteratively combining weak prediction models. However, similar to LSSVM, GBDT requires careful control of the number of iterations to prevent overfitting.

There are 32 papers discussing different neural networks, which are categorized into feedforward neural networks and feedback neural networks. ANN, as the most basic feedforward neural network, excels in approximating complex nonlinear relationships in static data and is one of the most popular models in CEPM research. BPNN, as a multilayer neural network, demonstrates good generalization ability and adaptivity; however, it has a slow convergence speed and is prone to local minima. In contrast, ELM uses the analytical solution method of randomly selecting hidden layer nodes, eliminating the need for iterative tuning and exhibiting efficient training speed on large datasets. However, ELM is not suitable for handling long-term carbon emission time series data, and its interpretability is relatively poor due to its stochastic nature. Feedback neural networks mainly include RNN and LSTM. RNN captures data features over time through its cyclic structure, but its hidden state relies heavily on the previous moment's output and current input, lacking a long-term memory advantage. LSTM, as an improved model of RNN, introduces gates and memory units, overcoming the limitations of traditional RNNs. This enables LSTM to perform well in predicting long-term carbon emission data, especially when considering economic changes and policy adjustments.

Combined prediction models include statistical-statistical models, statistical-intelligent models, and intelligent-intelligent models. Statistical-statistical models integrate classical statistical prediction models such as linear regression and exponential smoothing, offering advantages of simplicity, interpretability, and low complexity. They perform well in linear carbon emission datasets with clear relationships, mitigating errors through the combination of multiple base models. However, these models are less proficient in handling nonlinear and high-dimensional carbon emission data. Statistical-intelligent models combine classical statistical models with intelligent models like support vector machines and neural networks, capable of handling relatively complex carbon emission data. They leverage the strengths of both statistical and machine learning methods, suitable for a wide range of prediction tasks. The limitation lies in the determination of combination weights, which heavily relies on experience or other models. Intelligent-intelligent models, combining various shallow intelligent prediction models and neural networks, maximize prediction accuracy and stability. Nevertheless, these models exhibit high computational complexity, poor interpretability, and a susceptibility to overfitting. Due to their numerous parameters, optimization often requires the assistance of other models.

3.2. Optimization models

In CEPM, optimization is the process of finding the best possible solution to one or more models. As shown in Fig. 3, optimization models are broadly categorized into programming models and metaheuristic analysis. Metaheuristic analysis can be subdivided into four categories based on their intended purpose. As a result, the metaheuristic analysis used for parameter optimization accounts for more than half of these models (55 %). Among all the optimization models, the number of programming models is small, being used in only three papers. The Online Supplementary Appendix III shows the metaheuristic optimization results in CEPM in the research published during 2011–2022.

3.2.1. Programming models

Programming models originated from mathematical methods and are widely used in the field of model optimization. According to their structure, programming models can be divided into linear programming and nonlinear programming groups. In terms of optimization objectives, programming models can be divided into single-objective programming and multi-objective programming. In 2017 (Wang and Ye, 2017), Wang et al. introduced the power exponent of the relevant variable as an exogenous factor in the multivariate grey model. To minimize the mean absolute percentage error, they developed two nonlinear programming models, aimed at determining the power exponent for the nonlinear multivariate model.

3.2.2. Metaheuristic analysis (MA)

MA differs from traditional optimization methods by not relying on gradients, ensuring randomness and ease of use, while effectively bypassing local minima (Hussain et al., 2019). Typically, the optimization parameters of the MA model are defined initially, along with the objective function (Katebi et al., 2020). The search initiates with randomly generated candidate solutions, evolving to generate the subsequent generations. Within the 147 CEPM papers scrutinized, a predominant approach involves metaheuristic algorithms that emulate natural biological or physical phenomena, structuring mathematical models to tackle problems.

MA can be combined with predictive models, to rapidly select the most accurate parameters or parameter combinations, for the prediction model. Wen et al. (Wen and Cao, 2020b) proposed the enhanced butterfly optimization algorithm (EBOA) to optimize the parameters of an LSSVM. The results show that compared with the BOA-LSSVM, PSO-LSSVM, GA-LSSVM, CS-LSSVM and SVM models, the EBOA-LSSVM model yields the lowest error value. Sun et al. (Sun and Zhang, 2020) established the bacterial foraging optimization (BFO) method to optimize the regularization parameters and kernel parameters of LSSVM and used the subsequent model for prediction. The robustness and parameter optimization performance of the LSSVM were enhanced. Dai et al. (Dai et al., 2018) proposed the modified shuffled frog leaping algorithm (MSFLA) to optimize the regularization parameter and kernel function width of an LSSVM model. The prediction accuracy of the proposed MSFLA-LSSVM model was better than that of the considered SFLA-LSSVM, LSSVM and BPNN Models. Wu et al. (Wu et al., 2020) established a metaheuristic ant-lion optimizer (ALO) to optimize the fractional order α in the heterogeneous grey model CFNGM.

Structural optimization is defined as the optimization of the structure of a prediction model by using the performance characteristics of a metaheuristic model, such as the number of hidden layers in the neural network. The CS-PSO method was proposed by Chiroma et al. (Chiroma et al., 2015) to optimize the number of hidden layer neurons in ANN. Sun et al. (Sun et al., 2017) applied PSO to optimize the weight of the input layer and the deviation in the hidden layer in ELMs, thereby obtaining an optimal PSO-ELM network. The prediction accuracy was better than that of the traditional ELM model and BPNN model. Niu et al. (Niu et al., 2020) introduced the improved fireworks optimization algorithm (IFWA) to optimize the smoothing factor in GRNN, which

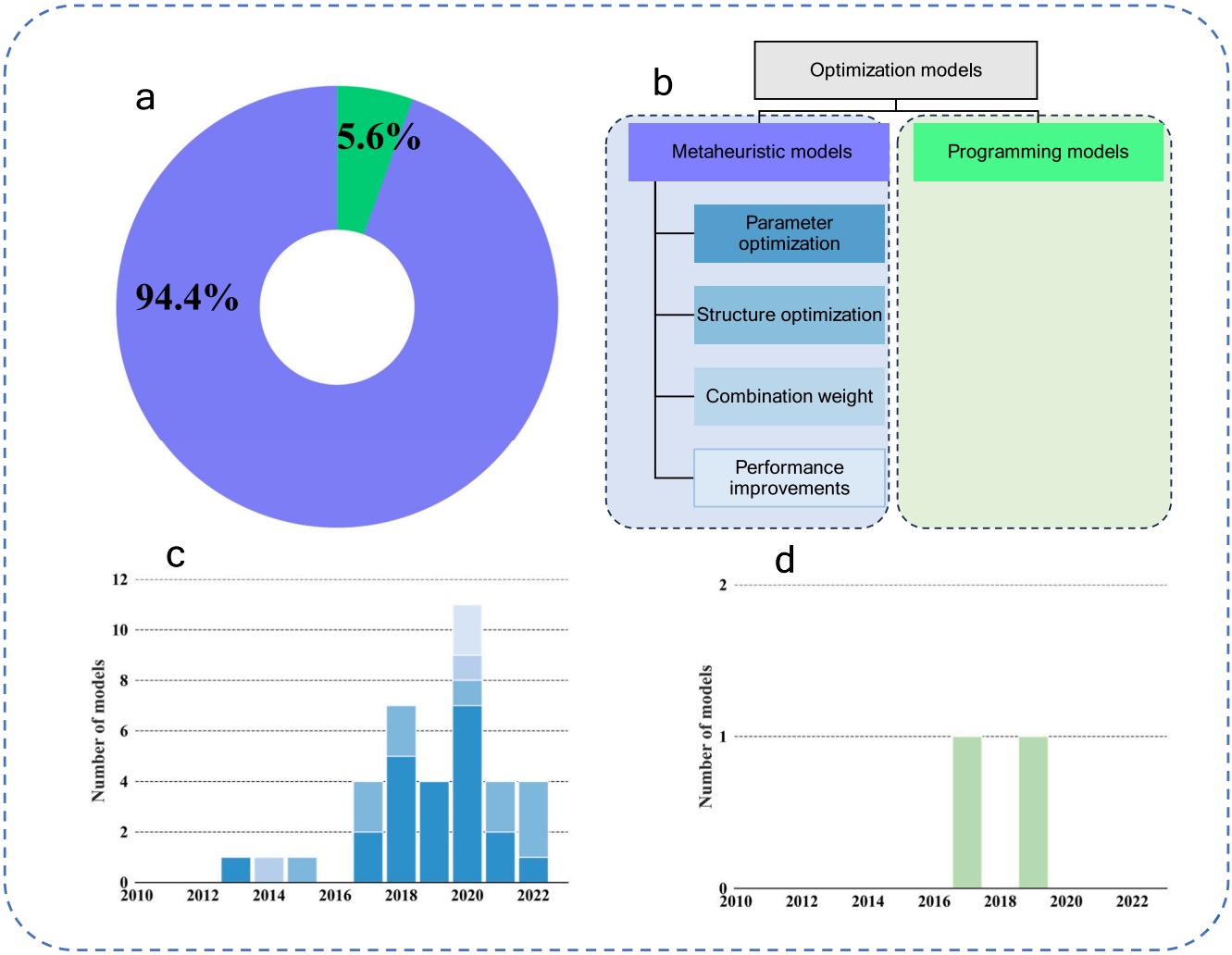


Fig. 3. (a) Proportion of the two optimization models among all models, with blue indicating mataheuristic models, green for programming models. (b) Classification of optimization models. (c) Time series distribution of different mataheuristic models, color-coded as shown in (b). (d) Time series distribution of different programming models.

improved the prediction accuracy. In the models reviewed, weight optimization was often the focus of structural optimization. A total of 9 papers (14, 51) applied weight optimization, and threshold optimization (Chai et al., 2022; Zhang et al., 2022) and learning rate optimization (Zhao et al., 2022) were also common.

There are many ways to construct a combined prediction model. The most important step is to ensure that the advantages of each prediction model are maximized. One such approach is to obtain the optimal weight coefficient combination for the applied prediction model. MA is used to search the optimal weights of each model to achieve the optimal prediction results for CEPMs. Liu et al. (Liu et al., 2014) constructed a combined prediction model with GM (1, 1), ARIMA and second-order polynomial regression (SOPR). PSO was used to optimize the combined weights of the above models, and the prediction accuracy was improved.

Performance improvements were discussed in some of the reviewed papers, especially in cases that involved optimizing shallow intelligent models. Wen et al. (Wen and Cao, 2020b) proposed the butterfly optimization algorithm (BOA) to optimize the performance of an LSSVM. Compared with the PSO-LSSVM, GA-LSSVM, CS-LSSVM and SVM models, the BOA-LSSVM model displayed the best performance. A similar performance improvement method was adopted by Wen et al. (Wen and Cao, 2020a), who used ICSO to improve the performance of a SVM.

3.2.3. *Summary of optimization models*

In conclusion, optimization models play a crucial role in CEPM, particularly in finding the optimal parameters for predictive models. This study categorizes optimization models into programming models and MA. Programming models, rooted in mathematical methods, are divided into linear and nonlinear programming groups, addressing single or multi-objective optimization. Metaheuristic Analysis is an advancement over traditional methods, utilizing algorithms inspired by natural phenomena. Among all the CEPM papers applying optimization models, MA is widely used, accounting for 94.4 %, with a focus on algorithms mimicking biological or physical processes. MA is applied for parameter optimization in predictive models. Examples include the Enhanced Butterfly Optimization Algorithm (EBOA) for optimizing LSSVM parameters, and the Ant-Lion Optimizer (ALO) for optimizing parameters in the grey model CFNGM. Structural optimization involves optimizing the structure of predictive models, such as the number of hidden layers in neural networks. Various Metaheuristic algorithms are employed for structural optimization, including CS-PSO for optimizing hidden layer neurons in ANN. Weight optimization is a common focus, with methods like PSO used to optimize weights in combined prediction models. Performance improvements were discussed in most of the reviewed papers, especially in cases that involved optimizing shallow intelligent models. In the majority of papers, optimization models enhance the performance and accuracy of predictive models in CEPM.

3.3. Prediction factor selection models

Selecting the most appropriate prediction factor for a prediction problem is a key step (Khaki and Wang, 2019). The traditional prediction model operates without preconceived assumptions about the distribution of the involved factors. However, carbon emission prediction is generally a macro-prediction task with many influencing factors (Cabaneros et al., 2019). Therefore, the robustness of a prediction model depends largely on the model form and the way prediction factors are input into the model. Additionally, prediction factor selection models can be used to extract the factors suitable for prediction, and the degree of correlation between different factor pairs can be determined. Prediction factor selection models can be divided into two categories, namely, statistical models and machine learning models, which are like prediction models. Some prediction models can also be used as prediction factor selection models, such as classical regression models. Moreover, machine learning models can be divided into supervised learning models and unsupervised learning models. The classification and quantitative of said models are given in Fig. 4.

3.3.1. Statistical models

In addition to forecasting, multiple linear regression serves as a pivotal tool for determining quantitative interdependencies among variables in prediction factor selection. Tian et al. (Tian et al., 2016) conducted an analysis on factors influencing carbon emissions in the crop industry using multiple linear regression, pinpointing social development and energy inputs as the predominant factors. Meanwhile, Zhao et al. (Zhao et al., 2022) proposed QAP regression analysis, and the results showed that the dominant factor leading to differences in carbon emissions in the Yellow River Basin from 2000 to 2010 was per capita GDP. After 2010, the main factor affecting carbon emissions in the Yellow River Basin was population. Ridge regression (RR) is another method for selecting the significantly correlated variables. RR can effectively avoid multicollinearity and is used by many scholars. The largest difference between RR and normal linear regression is the loss function. The regularization parameter lambda is introduced in RR to make the coefficients of unimportant prediction factors very small (Li, 2020). Using an RR model, Yu et al. (Yu et al., 2018) utilized an RR model to emphasize the notable impact of population aging on CO2 emissions. They highlighted a direct correlation between the aging population and the scale of CO2 emissions in their study.

Indeed, alongside grey prediction models like GM (1, 1) and GM (1, N), GRA holds significance within the grey system. It serves to gauge the strength and scale of relationships among system factors. Utilized for assessing connections between pairs of options through remote measurements, GRA assists in identifying quantitative correlations among intricate factors within a system. For small sample sets and irregular data, GRA has displayed good analytical performance in the field of

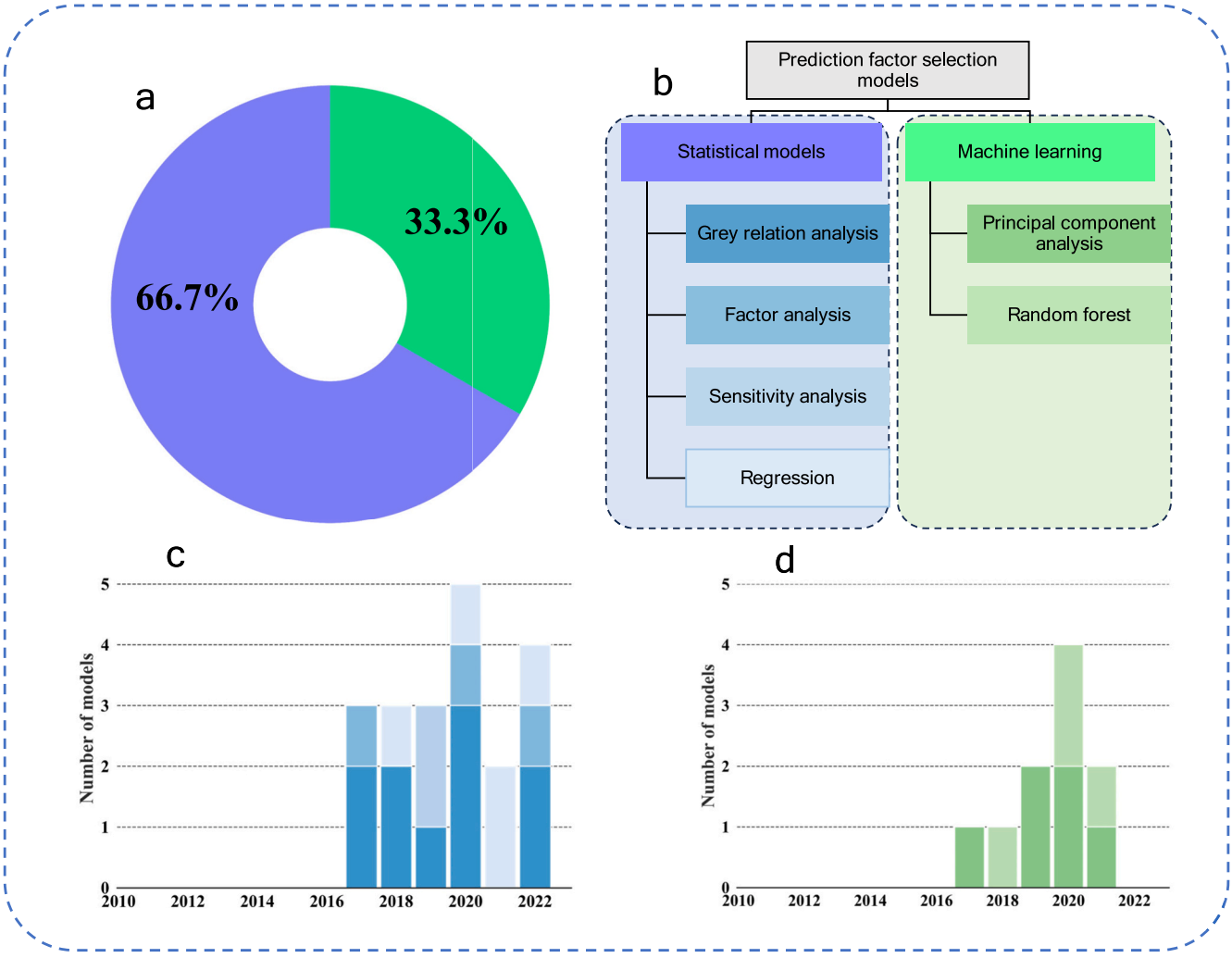


Fig. 4. (a) Proportion of the two prediction factor selection models among all models, with blue indicating statistical models, and green for machine learning models. (b) Classification of prediction factor selection models. (c) Time series distribution of different statistical models, color-coded as shown in (b). (d) Time series distribution of different machine learning models.

CEPM. Ding et al. (Ding et al., 2020) employed GRA to identify factors potentially exerting strong nonlinear influences on carbon emissions. This insight into nonlinear effects led to the design of a novel discrete grey model. Chiu et al. (Chiu et al., 2020) developed a new multivariate grey prediction model (MGPM) for CO2 emissions. GRA was used to filter the relevant features with weak correlations with carbon emissions and finally improve the prediction accuracy. Based on GRA, Zhou et al. (Zhou et al., 2017a) concluded that the factors that influence carbon emissions from strong to weak are the installed thermal power capacity, thermal power generation, urbanization rate, GDP and unit utilization.

FA is a statistical method employed to uncover underlying common factors within sets of variables (Jones et al., 2015). FA reduces large sets of prediction factors to a few factors by calculating the correlations between prediction factors. The extracted common factors serve to substitute the original prediction factors, preserving essential information and efficiently representing the intricate interrelations among the prediction factors. Sun et al. (Sun et al., 2017) integrated FA with an ELM for carbon emission forecasting, utilizing FA to identify crucial input prediction factors.

Sensitivity analysis (SA) assesses the influence of each input variable on carbon emission intensity by calculating the partial rank correlation coefficient (PRCC) between these variables. Acheampong et al. (Acheampong and Boateng, 2019) performed sensitivity analyses for carbon emissions in four countries. In Australia, R&D displayed the highest sensitivity weight. Urbanization exhibited the highest sensitivity weight in Brazil. Population size showed the highest sensitivity weight in China, and energy consumption demonstrated the highest sensitivity weight in India.

3.3.2. Machine learning

PCA is a prevalent method for dimensionality reduction and feature extraction in carbon prediction. It generates new orthogonal mappings to represent the data in a lower-dimensional space, capturing the maximum variance (Zhang et al., 2019). This reduction in dimensionality helps eliminate potential basic directions with limited information. PCA is commonly used to process and analyze various types of data, reducing high-dimensional and complex prediction factors to those suitable for analysis, and improving prediction quality. Scholars in CEPMs widely apply PCA in their studies (Wen and Cao, 2020a; Sun et al., 2019; Sun and Sun, 2017; Sun and Wang, 2021). For example, Huang et al. (Huang et al., 2019b) developed the grey-PCA-LSTM model. PCA is used for feature extraction, and 15 prediction factors were transformed into 4 prediction factors, ultimately improving the prediction accuracy of the model. Kernel principal component analysis (KPCA) was proposed by Wen et al. (Wen and Cao, 2020b). Initially, prediction factors with the most significant contributions to the selected function were extracted using KPCA. Subsequently, these prediction factors were combined with an EBOA-LSSVM prediction model to forecast carbon emissions in the residential sector of the Yangtze River Delta. The model with KPCA demonstrated superior performance compared to the original prediction model.

RF stands as an integrated machine learning approach widely employed in prediction and feature selection. Its strength lies in unraveling complex interactive features and quantifying feature importance. RF possesses robustness against noise, handles data gaps proficiently, and showcases swift learning abilities. Consequently, RF serves as a pivotal tool for feature selection, particularly in complex domains like CEPM. In their work Sun et al. (Sun et al., 2018) introduced a hybrid model merging RF and ELM techniques for CEPM. Remarkably, RF was utilized to scrutinize emission-influencing factors. Empirical simulations highlighted the superiority of the proposed RF-MFO-ELM model over parallel models like MFO-ELM, RF-PSOELM, RF-ELM, and RF-BP, showcasing substantial advantages in predictive fitting and accuracy.

3.3.3. Summary of the prediction factor selection models

In conclusion, the selection of prediction factors is a critical aspect in addressing macro-prediction tasks such as carbon emission prediction, characterized by numerous influencing factors. Prediction factor selection models fall into two main categories: statistical models and machine learning models. In the realm of statistical models, methods like MLR, RR, GRA, FA, and SA are prominent. MLR elucidates interdependencies between variables, RR effectively handles multicollinearity, GRA is suitable for small sample sets and irregular data to identify strong nonlinear influences, and SA assesses the impact of each input variable on carbon emission intensity. On the other hand, machine learning models such as PCA and RF provide alternative approaches. PCA achieves dimension reduction to enhance information while reducing complexity, and RF excels in revealing complex features and quantifying their importance. Despite the strengths of these models, they also have limitations. For instance, statistical models like MLR may struggle to capture complex nonlinear relationships, and the applicability of GRA may vary based on data characteristics. Machine learning-based selection models also have limitations; PCA may lose important information during dimension reduction, and RF, due to its relatively complex model structure, may pose challenges in interpretability.

4. Discussion

4.1. Influencing factors and research objectives

4.1.1. Influencing factors

The influential factors and the classification and quantitative statistics of influencing factors in CEPMs are shown in Fig. 5. All factors were divided into five sectors: energy (92 times), economy (111 times), population (68 times), environment (24 times) and industry (46 times).

Energy factors generally include energy consumption and energy structure, with 79 and 13 associated papers, respectively. Li et al. (Li et al., 2018b) extensively investigated carbon emission's primary energy sources in the Beijing-Tianjin-Hebei region, focusing on coal, gasoline, natural gas, and coal power. Their analysis emphasized the significant impact of energy consumption proportions on carbon emissions, foreseeing a 45 % projection in power and natural gas consumption by 2030. Additionally, Zhao et al. (Zhao et al., 2018b) scrutinized the multifaceted influences of GDP, population, energy consumption, economic structure, energy structure, urbanization rate, and energy intensity on carbon emissions. They revealed a negative correlation between the energy structure and carbon emissions.

Most studies have focused on the impacts of economic factors on carbon emissions. Economic factors can be divided into three categories: the overall economy, investment, and urbanization. In 67 papers, overall economic factors were used in CEPMs, although energy consumption factors were used in more studies. The main index used was GDP. Cao et al. (Cao et al., 2016) observed that GDP growth positively correlates with increased carbon emissions. In another study, Sun et al. (Sun et al., 2021) highlighted the positive influence of per capita GDP in Belt and Road countries on carbon emissions. Investment variables were also linked to carbon emissions, evidenced by Jiang et al. (Jiang et al., 2021a) identifying a nonlinear association between CO2 equivalent and foreign direct investment, with outward foreign direct investment amplifying CO2 emissions in China. Urbanization, the transition from traditional agricultural rural settings to modern urban societies, was identified as a significant factor affecting carbon emissions in China by Ma et al. (Ma et al., 2020) using the Apriori algorithm.

Population factors include the total population, policy, consumption level and education level. The total population is the most frequently used population factor. A total of 36 papers used the total population as an independent variable. Most papers were published in 2018, with a total of 9 papers discussing the impact of population on carbon emissions in that year. Cui et al. (Cui et al., 2018) explored the driving forces behind CO2 emissions in China's power industry, attributing increased

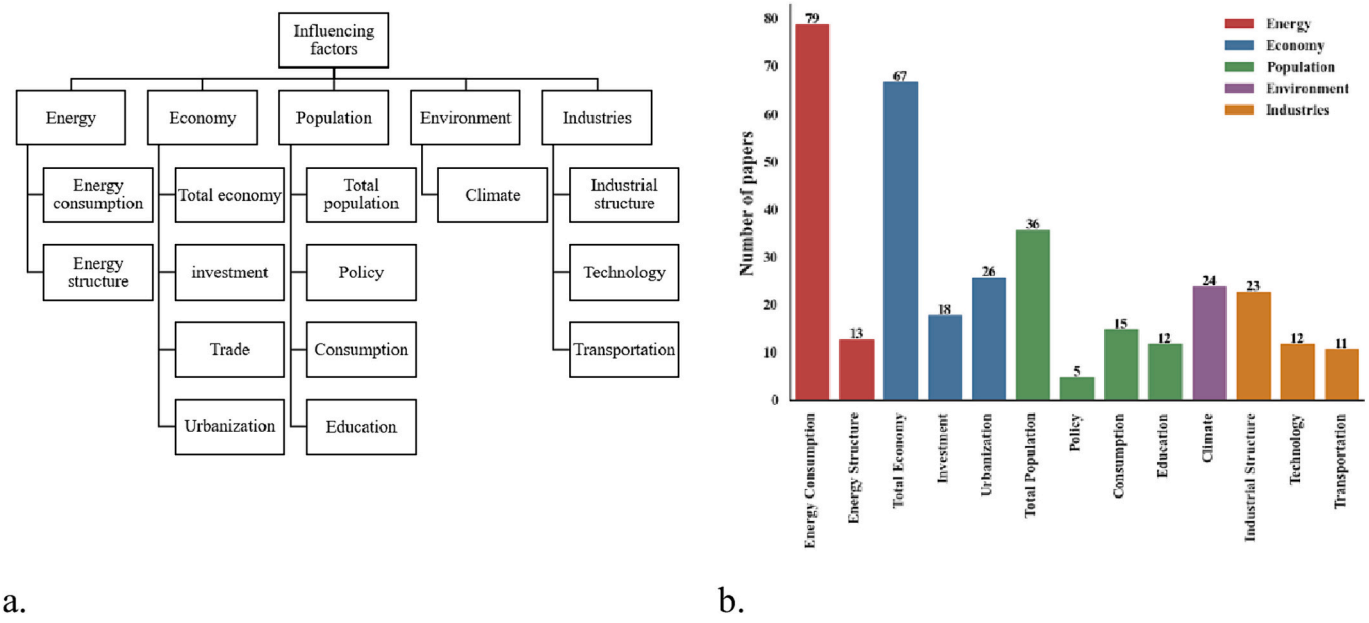


Fig. 5. (a) Classification of influencing factors. (b) Quantitative statistics of influencing factors.

emissions to human activities. Zhao et al. (Zhao et al., 2017), employing Grey Relational Analysis (GRA), revealed a robust correlation between China's population and carbon emissions, recording a correlation coefficient of 0.6798. Additionally, Niu et al. (Niu et al., 2020) conducted scenario analyses to explore China's commitment goals for 2030. They concluded that policy dynamics significantly influence carbon emissions, where policy-driven scenarios directly impact achieving these goals. The analysis by Sun et al. (Sun and Zhang, 2020) indicated that education plays a pivotal role as one of the main drivers of energy-related carbon emissions. Furthermore, consumption levels, another population factor, were discussed in 15 papers.

In 24 papers, the impact of environmental factors on carbon emissions was analyzed. Environmental factors, including climate factors, greenhouse gases (Yu et al., 2018; Sutthichaimethee and Kubaha, 2018; Tang et al., 2016), and temperature (Singh and Kennedy, 2015) were identified as the most commonly considered factors. Yu (Hao and Wei, 2015), after building an economic model, found that carbon efficiency is the most important factor in determining whether there is a turning point in total CO2 emissions. Singh and Kennedy (Singh and Kennedy, 2015) demonstrated, via scenario analysis, that climate change can have a marginal impact on the growth trajectory of energy-related carbon emissions.

In the reviewed papers, the main industry-related factors are the industrial structure, technological level, and transportation. Notably, Zhu et al. (Zhu et al., 2019) utilized a Markov chain model and scenario analysis to forecast industrial and energy structures. Their findings highlighted that adjusting the industrial structure could potentially have a greater impact on achieving carbon intensity targets than modifying the energy structure. Transportation-related factors involved metrics like the number of motor vehicles and transport mileage. Lin et al. (Lin et al., 2018) conducted a grey correlation analysis, revealing that the highest correlation with carbon emissions was observed with the number of registered motor vehicles. Additionally, the impact of scientific and technological progress on carbon emissions was explored in 12 studies.

4.1.2. *Research objectives*

The prediction models studied have different goals. In 147 papers, six targets were identified, including total population-related carbon emissions, economic sector carbon emissions, and industry carbon and energy-related carbon emissions. The classification and quantitative statistics of study targets are given in Fig. 6.

The overall targets include total carbon emissions (TCE) and the carbon emission intensity (CEI). There were 94 studies of total carbon

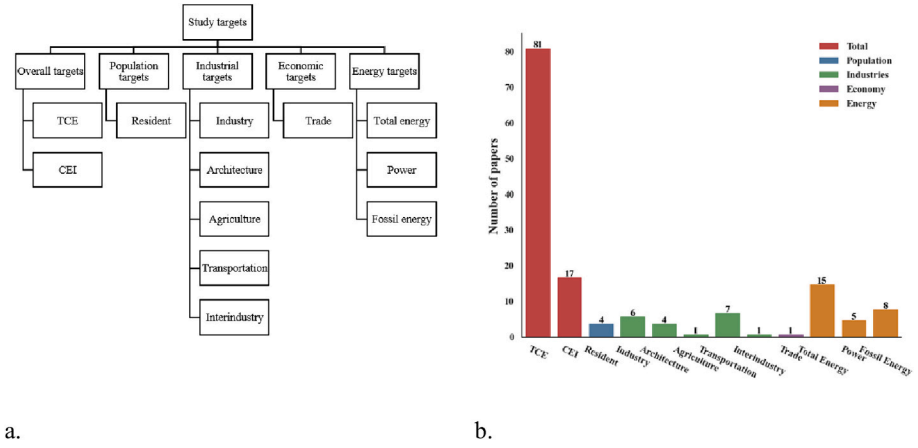


Fig. 6. (a) Classification of study targets. (b) Quantitative statistics of study targets.

emissions in the analysis, which is the most studied target, accounting for 63.5 % of all the papers. Additionally, there were far more TCE studies than CEI studies, and some studies focused on both TCE and the CEI. For example, Zhou et al. (Zhou et al., 2021) forecasted TCE and the CEI in China, and the results showed that both are projected to decrease from 2018 to 2030. Niu et al. (Niu et al., 2020) showed that in certain scenarios, China's TCE will gradually decline after reaching a peak in 2030. Additionally, there is still some pressure to achieve CEI reduction targets. Population-related carbon emissions are mainly carbon emissions in the residential sector. Four papers have studied carbon emissions in the residential sector. Wen et al. (Wen and Cao, 2020b) predicted the emissions from the residential sector in the Yangtze River Delta region and concluded that the carbon emissions from the residential sector increased from 55.86 Mt in 2000 to 192.36 Mt in 2017.

In the papers reviewed, the relevant carbon emissions from four sectors, namely, the industrial, agricultural, construction and transportation sectors, were analyzed. Among them, the transportation sector was most explored, with a total of seven papers discussing this sector in the context of carbon emissions. Huang et al. (Huang et al., 2022) studied carbon emissions from the transportation sector. Between 2019 and 2025, the average increases in carbon emissions from the transportation sector in China and the United States were 2.837 % and 2.394 %, respectively, and a decreasing trend was observed in Japan, with an average decline of 1.2231 %. Industrial sector carbon emissions were explored in a total of six papers. Yu et al. (Yu and Xu, 2019) used the GM (1, 1) model to conclude that industrial carbon emissions will peak in 2030, suggesting that the emission reduction targets will be met. In addition, the industrial sector includes the cement industry (Ofosu-Adarkwa et al., 2020) and the ferrous metal industry (Huang et al., 2019c). Two papers studied the steel industry in the ferrous metal industry (Gao et al., 2015; Li et al., 2022a). Carbon emissions in the construction industry were studied in four papers. Sutthichaimethee et al. (Sutthichaimethee and Kubaha, 2018) used a combined model to study the carbon emissions from the Thai construction industry over 20 years (2019–2038). The results showed that the carbon emissions from the construction industry will increase by 37.88 % in this period. In addition, one paper studied agricultural carbon emissions (Jiang et al., 2021b) and interindustry carbon emission transfer (Hu and Lv, 2020). One paper studied trade-related carbon emissions in the economic sector. Wang et al. (Wang et al., 2019b) delved into trade-related carbon emissions within the economic sector, simulating and predicting carbon emissions resulting from bilateral trade. Their findings indicated a significant increase in net carbon outflow from China to Australia, rising from 2.2 million tons to 15.5 million tons between 2000 and 2014. We divided energy-related carbon emissions into total energy carbon emissions, power-related carbon emissions and fossil energy carbon emissions. Fifteen studies focused on total energy carbon emissions, 5 papers focused on power-related carbon emissions, and 8 papers focused on fossil energy carbon emissions. Fossil energy carbon emissions include coal-related carbon emissions (Duan and Luo, 2020; Duan et al., 2020).

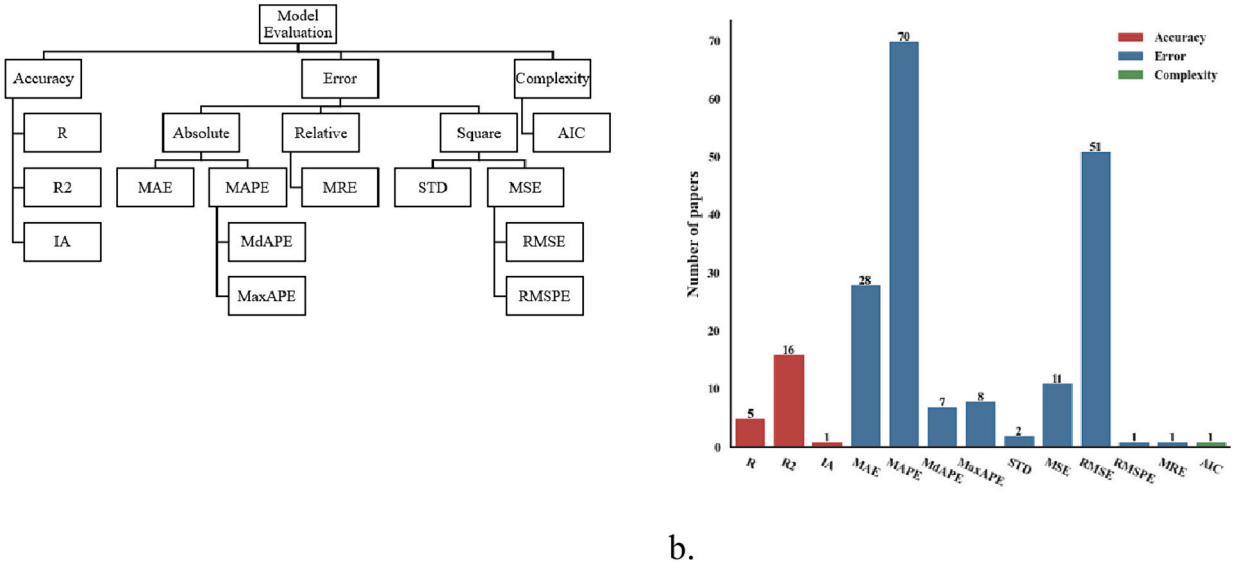
4.2. Model evaluation

4.2.1. Overview

Model evaluation methods are used to evaluate the performance of prediction models. The three main indicators include accuracy, error, and complexity. Among them, error is used the most frequently, and many derivative indicators have been applied, including the MAPE (70 papers), RMSE (51 papers) and MAE (28 papers). The specific classification and use frequency of these indexes are shown in Fig. 7. The evaluation results of prediction models in CEPM in the research published during 2011–2022 are shown in the Online Supplementary Appendix II.

A positive correlation exists between the accuracy index and the prediction accuracy of a model. In the 147 CEPM papers reviewed, three accuracy indexes were used: R, R2 and IA. The R value, also known as the Pearson correlation coefficient, was proposed by statistician Carl Pearson. R2, or the coefficient of determination, quantifies the proportion of the variance in the dependent variable y that's explained by the independent variable x in a regression model. This metric measures how well the model fits the observed data. Among the accuracy indicators, R2 is the most used (16 papers), followed by R (5 papers). In addition, Duan (Duan and Luo, 2020) used the index of agreement (IA) to evaluate the accuracy of a prediction model.

Unlike accuracy index, error index indicates good model performance when the corresponding values are small. CEPM research mainly involves absolute error, relative error, and square error. Absolute errors are based on the absolute difference between the actual and modeled outputs. Indicators of this type include the mean absolute percentage error (MAPE) and mean absolute error (MAE). Sun et al. (Sun and Zhang, 2020) used the maximum absolute percent error (MaxAPE) and median absolute percent error (MdAPE) to evaluate models. The square error is based on the square between the actual value and the model output. Common indicators include the standard error (STD) and mean square error (MSE). The commonly used root mean square error (RMSE) is an



a. b.

Fig. 7. (a). Classification and (b). Quantitative analysis of evaluation methods.

extension of MSE and was applied in 51 CEPM studies. Different from the absolute error, the traditional relative error refers to the ratio of the predicted value to the true value. Common examples of this type include the mean relative error (MRE), which was applied in three selected CEPM papers.

Complexity indicators include information standards, such as the Akaike information criterion (AIC). In addition to modeling error, the AIC considers the complexity of the model. Ning et al. (Ning et al., 2021) used the AIC to evaluate the ARIMA model. The results showed that the AIC values of the ARIMA (1,2,0) and ARIMA (2,2,0) models were 5.947830 and 5.643812, respectively.

4.2.2. Comparative analysis

As shown in Fig. 8, we counted the three most frequently used error evaluation metrics, RMSE, MAE, and MAPE. For comparison purposes, we focused solely on carbon emission studies with units in Mt. For the baseline CEPMs, the RMSE ranged from 0.11 to 1635 Mt. Among them, the Shallow Intelligent Prediction Model and the Ensemble Prediction Model performed well, with median RMSE values of 1.45 Mt and 0.42 Mt, respectively. Out of the nine Ensemble Prediction Models considered, eight had RMSE values <1 Mt, demonstrating stable performance. Statistical prediction models showed significant variability, with the highest and lowest RMSE models among all baseline CEPMs being statistical prediction models, specifically ARIMA (2, 1) (Ding et al., 2020) (1635 Mt) and EDGM (1, 1) (Guo et al., 2021) (0.11 Mt). This indicates the need for improved stability in statistical models. For neural network prediction models, the RMSE ranged from 0.30 to 212.51 Mt. Some studies evaluated the MAE error of the models, with the MAE for all models distributed between 0.09 and 452.57 Mt. Comparing neural network prediction models and statistical prediction models, their respective median MAE values were 0.45 Mt and 104.22 Mt. However, some statistical prediction models had large MAE errors, with six out of thirteen models exceeding the maximum value of the neural network prediction model (163.68 Mt). All models were evaluated using MAPE, and the MAPE for all models ranged from 0.008 % to 37.14 %.

Further analysis of the optimized model errors showed a reduction in RMSE for all models, with the minimum and maximum values being 0.0003 Mt and 95.14 Mt, corresponding to the WOA-ELM (Sun and Huang, 2022) and BA-BPNN (Li et al., 2022a) models, respectively. The MAE for the optimized models also decreased, with median values of 0.24 Mt for neural network prediction models and 30.9 Mt for statistical prediction models. For MAPE, the optimized models showed a significant decrease, ranging from 0.002 % to 6.54 %.

4.3. Temporal and spatial distributions

4.3.1. Temporal distribution

a. Overall temporal trend

Fig. 9 shows the temporal distribution of review papers and the utilization frequency of prediction models in papers by year. The first prediction model was published in 2011. Initially, a few models were developed, with 2, 3, 1 and 3 in 2011, 2012, 2013 and 2014, respectively. The number of papers involving CEPM was 7 in 2015, increasing to 6 and 10 in 2016 and 2017, respectively. Perhaps affected by the coronavirus epidemic, the number of research papers decreased from 18 to 12 from 2018 to 2019. In the past five years, 78.4 % of all the CEPM papers included in this study were published. From 2019 to 2020, the number of CEPM increased from 12 to 26, an increase of 115.4 %. Similarly, in 2020, China proposed a carbon neutrality plan, and there may be a certain correlation between this plan and the publication trend. In the next year, 2021, the largest number of papers (31 papers) was published. The scope of this review extended through August 2022,

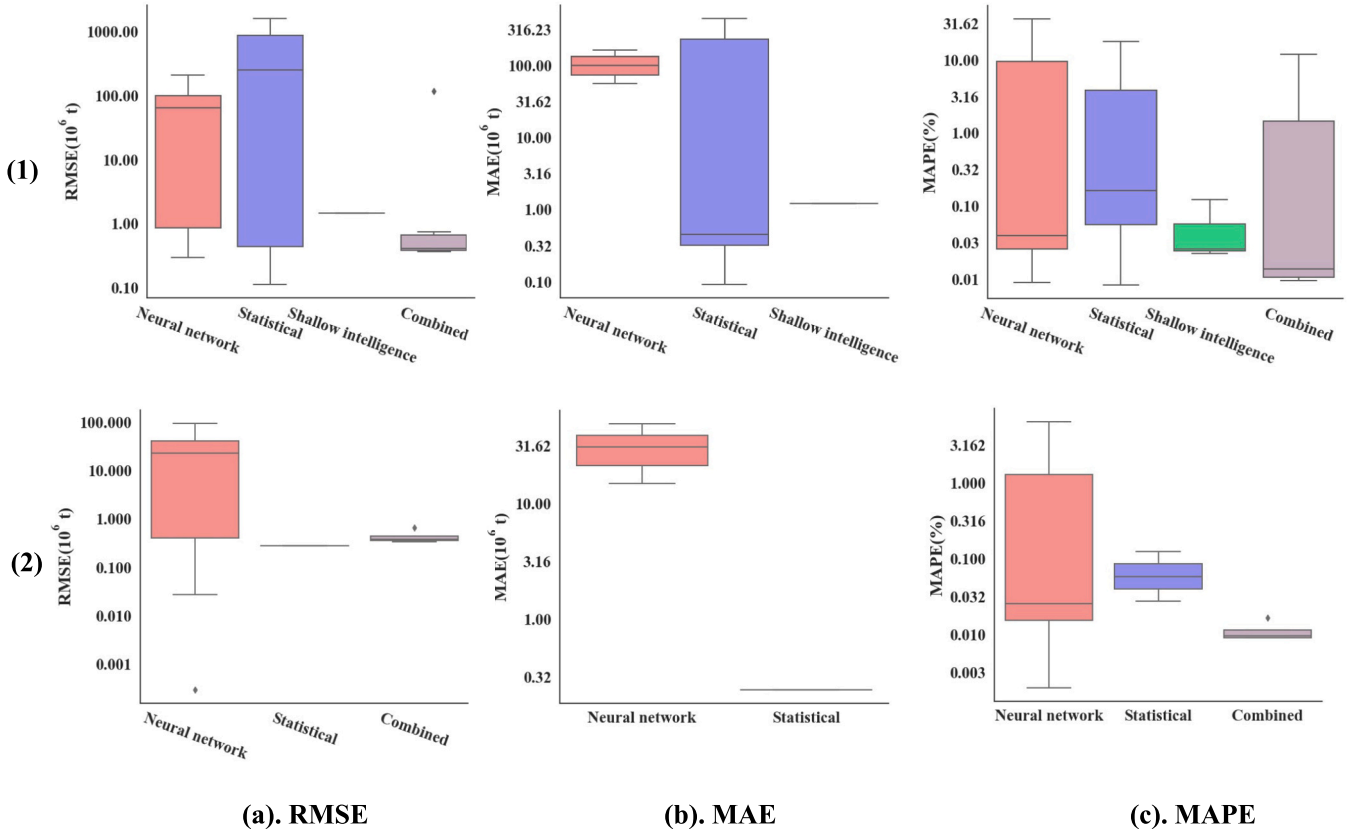


Fig. 8. (1) Accuracy of baseline CEPMs. (2) Accuracy of optimized CEPMs.

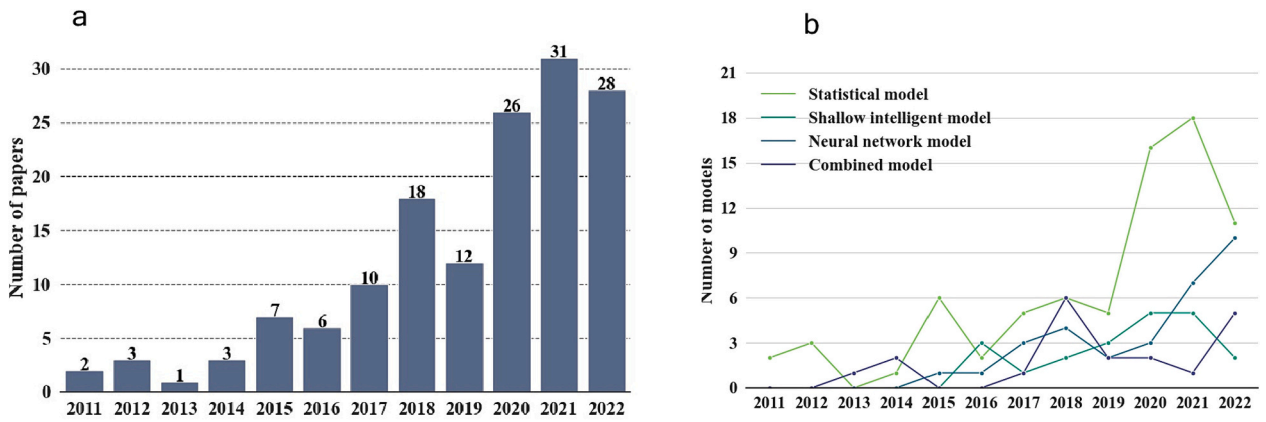


Fig. 9. (a) Temporal distribution of review papers. (b) Utilization frequency of prediction models in papers by year.

so the number of papers on CEPM is likely to continue to rise in 2022, and even for the foreseeable future.

b. Model comparison

Among the forecasting methods, the first to be used in the study period were statistical models, of which GM was the first (Pao and Tsai, 2011). Since 2014, statistical models have been used every year, with an overall upward trend peaking in 2021, with a total of 18 papers that used statistical models. Since 2015, the use of shallow intelligent models and neural network models has increased significantly. Shallow intelligent models reached a peak (5 papers) in 2020 and 2021 and have since declined in use, while neural network models have continued to gradually increase in use, with 3, 7, and 10 corresponding papers published in 2020, 2021 and 2022, respectively. After 2017, combined prediction methods began to increase in popularity. However, combined model use displays no obvious trend, and the fluctuation range is large. In 2018, 6 papers used combined prediction methods, making these models 4 times more popular than shallow intelligent models (2 papers). 2019–2021, the utilization frequency decreases to less than four for combined models. The reason for this trend may be that combined models are more computationally complex and take longer to run than individual models.

4.3.2. Spatial distribution

As shown in Fig. 10a, 83 countries were investigated 282 times in the studied papers. Most of the papers focused on the carbon emissions of individual countries, and 12 papers considered carbon emissions in multiple countries, including those from interstate organizations, on five continents. In addition, 7 papers focused on global carbon emissions, and in 1 paper, carbon emissions throughout Europe were predicted.

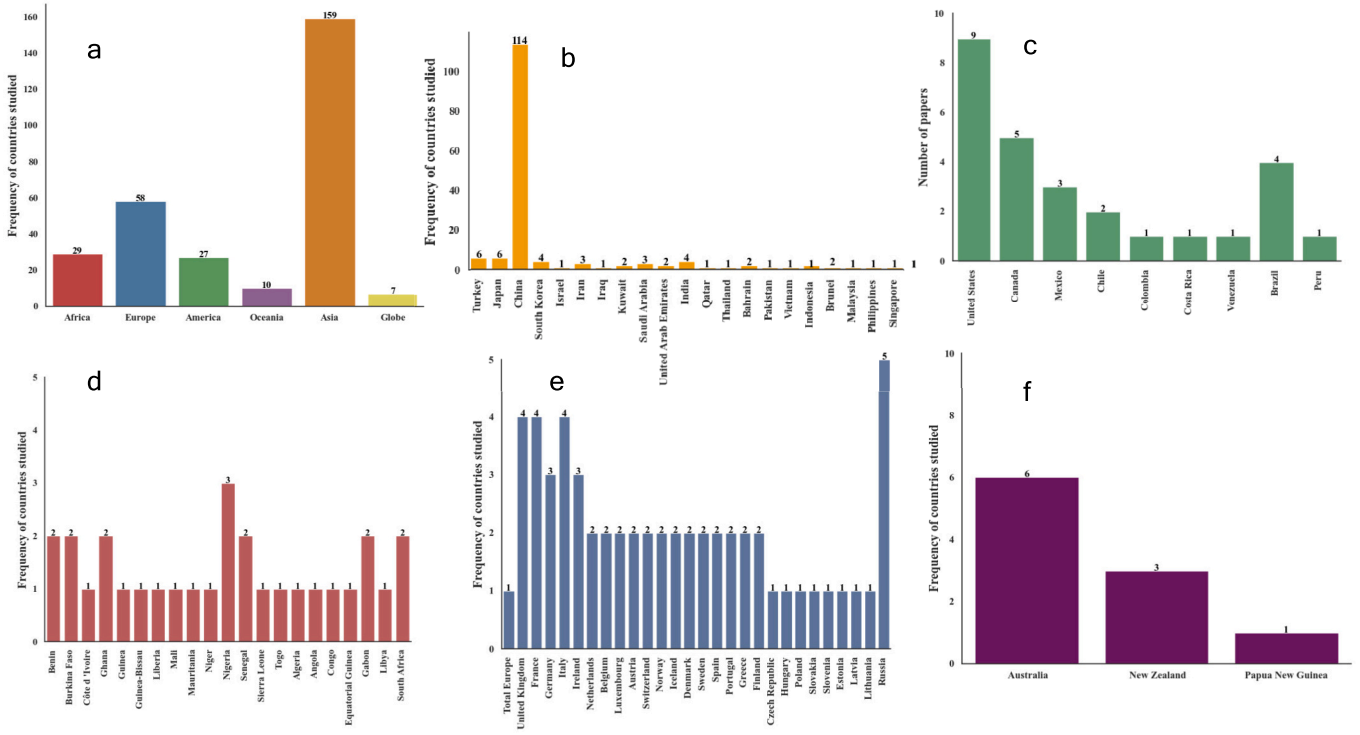


Fig. 10. (a). Frequency statistics of countries studied on five continents. (b). Frequency statistics for the Asian countries studied. (c). Frequency statistics for the American countries studied. (d). Frequency statistics for the African countries studied. (e). Frequency statistics for the European countries studied. (f). Frequency statistics for the Oceanian countries studied.

Among the five continents, Asia was most studied, and 22 countries were mentioned 159 times, with 114 in reference to China. Among all the papers discussing China, 38.6 % studied different regions of China, including cross-provincial regions (6 papers), provincial regions (33 papers) and municipal regions (5 papers). In addition to China, Turkey and Japan were studied 6 times each, followed by South Korea and India at 4 times each. The frequency statistics for the Asian countries studied is given in Fig. 10b.

As shown in Fig. 13c., a total of 9 countries in the Americas were studied, of which the United States was the most frequently studied country in North America, with a total of 9 papers, followed by Canada (5 papers). In South America, Brazil was most studied (4 papers), mainly because Brazil is a member of the BRICS organization, which was studied three times.

Oceania displayed the lowest number of countries studied at 3, namely, Australia (6 papers), New Zealand (3 papers) and Papua New Guinea (1 paper). The frequency statistics for the Oceanian countries studied are given in Fig. 10 f.

As shown in Fig. 10e, Europe was second only to Asia in terms of mentions across all papers reviewed, with 58 mentions across 27 countries and Europe as a whole. In terms of the frequency distribution, the countries displayed relatively similar trends, with 8 countries studied once and 13 countries studied twice. Russia was the most studied country (5 papers), followed by the UK, France and Italy (4 papers each).

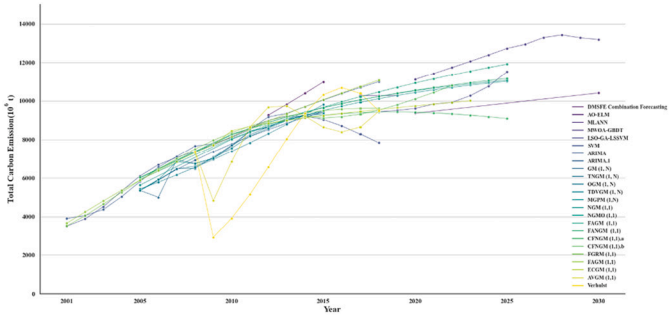
As shown in Fig. 10d, Similar to the findings for Europe, the frequency of study of African countries was relatively consistent, ranging from 1 to 3 times. Most countries were studied once, with only Nigeria being studied three times.

4.4. Future trends in carbon emissions

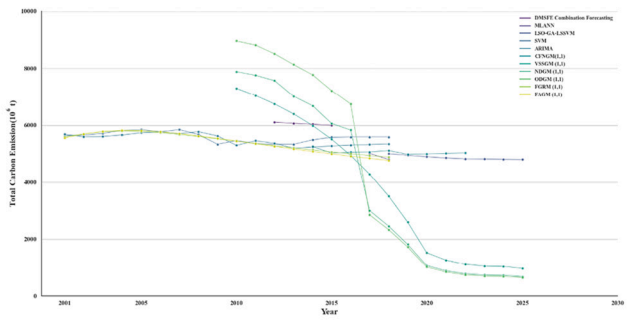
In this section, we examine the countries with large numbers of studies and their future carbon emissions predicted with CEPMs. The four selected countries are China, the USA, Japan, and Russia. As shown in Fig. 11, according to the previous model classification, we rank the carbon emission prediction values. Models with high levels of intelligence are shown in darker colors, and shallow intelligent models and statistical models are illustrated in light colors.

4.4.1. China

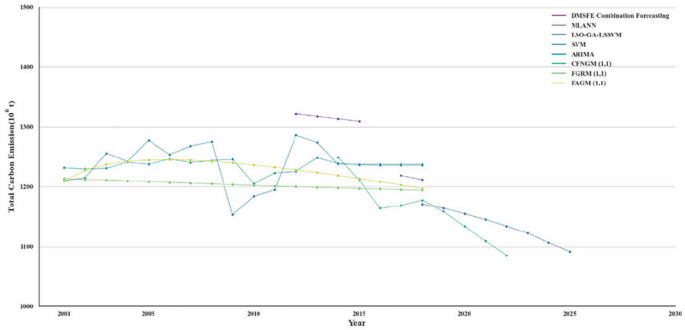
Fig. 11a shows the future predictions of carbon emissions in China in review papers. Most of the models provide consistent assessments of future carbon emissions in China, with carbon emissions likely to increase through 2025 or 2030. Notably, the MWOA-GBDT forecast for 2020–2030 is above average. It is worth noting that a SVM model prediction suggests that carbon emissions in China peaked in 2013 and declined year by year from 2013 to 2018. The predicted value in 2018 was 7841.6 Mt, which is lower than the predicted value of all other models in 2018. The maximum value in 2018 was 11,081.9 Mt predicted by FAGM (1, 1), which is 41.3 % higher than the prediction of a SVM. In addition, the Verhulst model predictions fluctuate significantly from 2008 to 2018. AVGM (1, 1) also displays similar fluctuations. Overall, for CEPM in China, highly intelligent models tend to yield higher predicted values, but overall fluctuations are stable. In addition, researchers prefer to use intelligent prediction models to forecast large-span time series. In contrast, shallow intelligent models and statistical models, especially various types of GMs, have displayed good prediction performance; thus, GMs are the most widely used prediction models. However, models with low intelligence are prone to large fluctuations in predicted values.



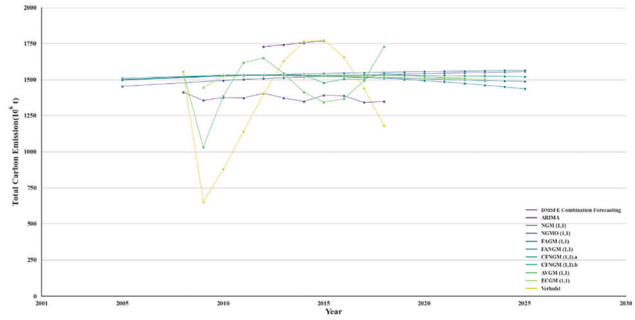
a. China[43, 117, 143-152]



b. USA[143-147, 149]



c. Japan[143-145, 147, 149]



d. Russia[43, 144, 145, 148, 151]

Fig. 11. (a). Future predictions of carbon emissions in China in review papers (Wu et al., 2020; Chiu et al., 2020; Qiao et al., 2020; Chang et al., 2013; Xu et al., 2021; Ikram et al., 2021; Gao et al., 2021; Tong et al., 2021; Huang et al., 2021; Qiao et al., 2021; Gao et al., 2022; Li et al., 2022b). (b). Future predictions of carbon emissions in the Americas in review papers (Qiao et al., 2020; Chang et al., 2013; Xu et al., 2021; Ikram et al., 2021; Gao et al., 2021; Huang et al., 2021). (c). Future predictions of carbon emissions in Japan in review papers (Qiao et al., 2020; Chang et al., 2013; Xu et al., 2021; Gao et al., 2021; Huang et al., 2021). (d). Future predictions of carbon emissions in Russia in review papers (Wu et al., 2020; Chang et al., 2013; Xu et al., 2021; Tong et al., 2021; Gao et al., 2022).

4.4.2. USA

As shown in Fig. 11b, we found that 11 public prediction models were reported. Different models displayed obvious differences. Three models predicted that the carbon emissions of the United States would plummet in 2016 to 2017 and will approach zero in 2025. The predicted values of VSSGM (1, 1), NDGM (1, 1) and ODGM (1, 1) for the United States in 2025 are 965.5855, 675.90985 and 642.1143575, respectively, which are much lower than those of other models. Among the other eight models, MLANN has the shortest prediction span of two years. In addition, the predicted trends of 8 models are stable, with 3 models indicating a slow increasing trend and 5 models indicating a slow decreasing trend.

4.4.3. Japan

Among the four major countries reviewed, Japan has the lowest forecasted values of total carbon emissions, with an average value of 1215.66 Mt. The maximum value predicted by the combined DMSFE model for 2012 was 1321.4904 Mt. The two minimum values are predicted after 2020: 1085.31 Mt in 2022 predicted by CFNGM (1, 1) and 1091.78 Mt in 2025 predicted by the LSO-GA-LSSVM. The future predictions of carbon emissions in Japan in review papers is given in Fig. 11c.

4.4.4. Russia

As shown in Fig. 11d, the overall future predictions for Russia are like those for China, with 18.2 % of the models, such as the Verhulst model and the AVGM (1, 1) model, displaying significant fluctuations. A total of 45.5 % of the model prediction curves are similar, and the range of carbon emissions is between 1400 and 1600 Mt. The corresponding models are NGM (1, 1), NGMO (1, 1), FAGM (1, 1), FANGM (1, 1), and CFNGM (1, 1).

4.5. Limitations of existing research and future suggestions

Firstly, we should consider the influence of significant events, such as COVID-19, in the CEPM domain. The novel coronavirus pandemic has had a profound impact on various fields of research around the world. However, few studies have considered the impact of novel coronavirus epidemic factors, such as the number of novel coronavirus infections or mortality, on future carbon emissions. Secondly, some studies have explored multiple regions in one paper, but only individual regional predictions were provided, without considering the geographical links among regions. Like other gases in the atmosphere, CO_2 can be transported, so the links between carbon emissions and geographic and other factors need to be further considered. Addressing this issue requires additional scientific data collection and in-depth data analysis. Thirdly, Machine learning techniques, including shallow intelligent models and neural network models, have been proven to be suitable for big data forecasting and have extremely high accuracy, including in stock forecasting and building energy consumption forecasting (energy consumption forecasting 1). However, in the field of carbon emissions forecasting, there are not significantly more studies that use machine learning techniques than statistical models. The reason for this phenomenon may be that most of the carbon emissions forecasting data are annual data (134 papers), so the amount of data is small, and it is difficult to fully train machine learning models. Determining how to overcome the inherent disadvantages of the data is worth studying in the future. Finally, all the prediction models reviewed predict carbon emissions at the city level, provincial level, cross-provincial level, national level and global level to help control carbon emissions at the macroscales, but these models are difficult to implement at the building level. Due to the lack of research on specific buildings, large amounts of data are not available to statistically analyze, undoubtedly hindering the development of CEPM.

5. Conclusion

To explore CEPMs, we summarize 147 relevant papers published between 2012 and August 2022 and classify the models into three main groups: prediction models, optimization models, and prediction factor selection models. In addition, we discuss other indispensable components of CEPM, such as influencing factors, research objectives, model evaluation, spatial and temporal scales, and future trends. We draw the following conclusions from this review.

Among all the prediction models, statistical models were the most prevalent, accounting for 75 instances—more than half—followed by neural network models at 21.8 %. Notably, the use of neural network models has consistently risen from 2019 to 2022, predominantly favoring feedforward architectures. A substantial portion of CEPMs incorporated optimized approaches, with metaheuristic models constituting 94.4 %. Within this category, parameter optimization took precedence, followed by structure optimization. Prediction factor selection models played a role in filtering factors, employing Grey Relational Analysis (GRA) and Principal Component Analysis (PCA) as representatives for statistical and machine learning models, respectively.

We scrutinized the accuracy of both pre-optimized and post-optimized CEPMs. In the pre-optimized stage, RMSE values spanned from 0.112 to 1635 Mt, showcasing a polarized performance among statistical prediction models—some highly accurate, others less so. Combined prediction models, however, displayed a relatively consistent performance. Post-optimization, there was a noticeable improvement across the board, with RMSE, MAE, and MAPE values reducing. The minimum RMSE reached 0.0003 Mt, and the maximum was 95.14 Mt.

CEPMs typically factored in five influences: energy, economy, population, environment, and industry. Energy and economic considerations dominated, with 92 and 111 times, respectively.

Drawing insights from the reviewed literature, we outlined prospective carbon emission trajectories for China, the United States, Japan, and Russia. China exhibited a continuous upward trend until 2025 in six models, with only one suggesting a minor decline. Projections indicate China's total carbon emissions may surpass 10,000 Mt by 2025. Conversely, the majority of models for the United States indicated a decline, with certain results suggesting the U.S. might approach carbon neutrality by 2025, as suggested by VSSGM (1, 1), NDGM (1, 1), and ODGM (1, 1).

CRediT authorship contribution statement

Yukai Jin: Writing – original draft, Formal analysis, Conceptualization. **Ayyoob Sharifi:** Writing – review & editing, Supervision. **Zhisheng Li:** Writing – review & editing, Formal analysis. **Sirui Chen:** Writing – review & editing, Formal analysis. **Suzhen Zeng:** Writing – review & editing, Data curation. **Shanlun Zhao:** Writing – review & editing.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used AI-assisted tools to check the grammar, punctuation, and clarity. After using the tools, the author(s) reviewed and edited the content as needed and take (s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A

List of abbreviations, including units and nomenclature.

Abbreviations	Nomenclature
CEPM	CO2 emission prediction model
TCE	Total carbon emissions
CEI	Carbon emission intensity
GM	Grey model
PR	Panel regression
LR	Linear regression
GPR	Gaussian process regression
EKC	Environmental Kuznets curve
NGBM	Nonlinear grey Bernoulli model
ARMA	Autoregressive moving average
ARIMA	Autoregressive integrated moving average
ARDL	Autoregressive distributed lag
SVM	Support vector machines
EKC	Environmental Kuznets curve
ELC	Environmental logic curve
GSM	Green-Solow model
BPNN	Backpropagation neural network
GPR	Gaussian process regression
LSSVM	Least squares support vector machine
GBDT	Gradient boosting decision tree
CART	Classification and regression tree
NN	Neural network
FNN	Feedforward neural networks
ANN	Artificial neural network
NNAR	Neural network autoregressive
GRNN	Generalized Regression Neural Network
RBFNN	Radial Basis Function Neural Network
CNN	Convolutional neural network
ELM	Extreme learning machine
RNN	Recurrent neural network
LSTM	Long short-term memory
BiLSTM	Bidirectional long short-term memory
LMDI	Logarithmic mean Divisia index
ROS	Random oscillation sequence
GVM	General vector machine
MIDAS	Mixed frequency data sampling
OLS	Ordinary least squares
PDL	Polynomial distribution lag
ADL	Autoregressive distribution lag
DMSE	Discounted mean square error
QHS	Quantum harmony search
KF	Kalman filter
MA	Metaheuristic analysis
AFSA	Artificial fish swarm algorithm
KF	Kalman filter
BOA	Butterfly optimization algorithm
EBOA	Enhanced butterfly optimization algorithm
PSO	Particle swarm optimization
BFO	Bacterial foraging optimization
MSFLA	Modified shuffled frog leaping algorithm
CS	Cuckoo Search
ALO	Ant-lion optimizer
FWA	Fireworks optimization algorithm
IFWA	Improved fireworks optimization algorithm
SOPR	Second-order polynomial regression
GRA	Grey relation analysis
SA	Sensitivity analysis
FA	Factor analysis
PCA	Principal component analysis
RELM	Regularized extreme learning machine
RF	Random Forest
LCA	Life cycle assessment
R^2	The coefficient of determination

(*continued on next page*)

(*continued*)

Abbreviations	Nomenclature
IA	The index of agreement
MAPE	Mean absolute percentage error
MAE	Mean absolute error
MaxAPE	Maximum absolute percent error
MdAPE	Median absolute percent error
STD	Standard error
RMSE	Root mean square error
MRE	Mean relative error
AIC	Akaike information criterion
IPCC	Intergovernmental Panel on Climate Change

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2024.172319.

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